

Keeper tier-gate — Lyra bulk-color v0.5 (Toeplitz setup + 8 = 3+3+2 count match) + Quasi-Eigentone v0.2 (Resolution B: T190 exponent = gauge mediator). Bulk-color v0.5: CANDIDATE COUNT MATCH at IDENTIFIED tier with explicit substantive concern on the SO(3)-branching mismatch Lyra herself caught (su(3) under SO(3) = 3+5, NOT 3+3+2). Quasi-Eigentone v0.2: PASS at SYNTHESIS tier with explicit two-structures-burden flag on Resolution B (one C_2=6 playing two roles parallels the #414 v0.2 reframe pattern).

Keeper

2026-05-30 Saturday late-morning batch

Tier-gate — bulk-color v0.5 + Quasi-Eigentone v0.2

1. Lyra bulk-color v0.5 — Toeplitz setup + 8 = 3+3+2 count match

Verdict: CANDIDATE COUNT MATCH at IDENTIFIED tier with explicit Cal #27 brake on the structural-vs-coincidence question + one substantive concern Lyra herself flagged that needs explicit handling in v0.6.

Strengths:

- Explicit Toeplitz setup is rigorous: $T_f = P_+ M_f$ on $H^2(S)$; commutator ideal = compact operators $K(H^2(S))$; short exact sequence $0 \rightarrow K \rightarrow T(H^2(S)) \xrightarrow{\wedge \sigma} C(S) \rightarrow 0$. The Upmeyer/Vasilevski symbol calculus framework is real established math (Cal #33 sourcing condition C1 from my v0.4 tier-gate carries forward).
- Paradox resolution: non-abelian structure from abelian p_+ is genuinely resolved via Hardy-projection-induced compact commutators. The substrate's $[p_+, p_+] = 0$ obstruction (Elie A1) does NOT preclude non-commutative Toeplitz algebra. Structurally clean.
- SO(3) sub-vector $\rightarrow$ 3 K-equivariant $T_a + 3$ pairwise $[T_a, T_b] \in K + 2$ Cartan from K rank-2 = 8 — first numerical count alignment for SU(3) 8-gluon emergence.
- Lyra's own Cal #27 self-application is exemplary:
 - “8 = 3 + 3 + 2 is one of MANY ways to decompose 8”
 - “The construction is GENERIC enough that it would work for any rank-2 group acting on a Hardy space with a 3-direction sub-structure — not unique to su(3)”
 - “Per Cal #27: ‘candidate count match’ with explicit ‘structural closure pending’”
- Internal self-correction in Section 2.4 — Lyra noticed her su(3)-under-SO(3) analogy was wrong (“3 (vector) + 5 (symmetric traceless)... wait, that's a different decomposition”) and flagged it explicitly. This is the right discipline.

Substantive concern requiring explicit v0.6 handling (the issue Lyra self-flagged):

- The 3+3+2 assembly does NOT match su(3)'s SO(3)-graded decomposition. Standard $SU(3) \supset SO(3)$

branching: adjoint $8 = 3$ (vector) + 5 (symmetric traceless), not $3+3+2$. So the proposed $(3 T_a + 3 [T_a, T_b] + 2 \text{ Cartan})$ doesn't sit in $\mathfrak{su}(3)$ in the canonical $\text{SO}(3)$ -equivariant basis Lyra was implicitly invoking. Three options for v0.6:

- (a) The substrate's $\mathfrak{su}(3)$ emerges with a NON-CANONICAL $\text{SO}(3)$ embedding (i.e., the 3 $\text{SO}(3)$ labels aren't the vector-3 of $\mathfrak{su}(3)$'s $\text{SO}(3)$ subgroup but some other rank-2 sub-structure inside K). Needs explicit identification.
- (b) The $3+3+2$ IS $\mathfrak{su}(3)$ in a different decomposition than canonical $\text{SO}(3)$ -graded (perhaps Cartan-graded under $\text{SO}(3) \times \text{U}(1)$ instead). Needs explicit structure-constant verification.
- (c) The count-8 is pattern-match coincidence; the construction doesn't actually close into $\mathfrak{su}(3)$. Honest possibility.
- The closure verification (whether $\{T_a, [T_a, T_b], \text{Cartan}\}$ satisfy $\mathfrak{su}(3)$ commutation relations) is the multi-week determination. Without it, v0.5 is COUNT ALIGNMENT, not derivation.

Conditions for v0.6 (joint with previous v0.4 conditions C1+C2):

- C1 (carried from v0.4): Toeplitz-source citation (Upmeyer *Toeplitz Operators and Index Theory in Several Complex Variables* or Vasilevski) — VERIFIED-CITED with theorem/page for the load-bearing claim “ $T(H^2(S))$ non-commutative even when boundary multiplication abelian” applied to Shilov boundary of D_{IV}^5 .
- C2 (carried from v0.4): AdS/CFT analogy demoted to evocative parallel only.
- C3 (NEW): explicit treatment of the $\text{SO}(3)$ -decomposition-mismatch ($3+3+2$ vs canonical $3+5$). v0.6 must either (a) identify the non-canonical embedding, (b) verify a non-canonical $\mathfrak{su}(3)$ structure, or (c) honestly mark the construction as “count-aligned, structurally OPEN.”
- C4 (NEW): explicit verification (Elie's lane) that $[T_a, T_b]$ are LINEARLY INDEPENDENT — the 3 pair-wise commutators are claimed independent but not verified.

Tier disposition: - Toeplitz-algebra setup (Section 1): RIGOROUS modulo C1 sourcing - $\text{SO}(3)$ sub-vector instantiation (Section 2.1-2.3): CANDIDATE structural identification - Count $3+3+2=8$ (Section 2.4): IDENTIFIED count match; STRUCTURAL CLOSURE PENDING per Lyra's own honest scope - $\text{SU}(3)$ closure verification: OPEN multi-week (C3+C4 conditions) - QCD low-energy phenomenology: OPEN multi-week

Status: PASS at CANDIDATE COUNT MATCH; the headline “first numerical count match for 8-gluon emergence” is correct as stated by Lyra; the C3 $\text{SO}(3)$ -decomposition concern is the substantive next-step that v0.6 must address. Not promoted to DERIVED; closure verification carries the burden.

Headline-cap-conditionality check (Calibration #34 STANDING): Lyra's headline already correctly carries “candidate-count-alignment; structural verification multi-week” — passes Cal #34 cleanly.

2. Lyra Quasi-Eigentone Framework v0.2 — Resolution B

Verdict: PASS at SYNTHESIS tier. Cleanly absorbs three Saturday-team pieces (Grace 1.5 Pair α channels + Elie Toy 3627 radial towers + L4 v0.2 T190 tension) into a refined framework. Resolution B is structurally well-motivated; one important “two-structures burden” flag carries forward.

Strengths:

- Three-piece synthesis is cross-CI cohesion at its best: Grace's intermediate-Casimir channel labels + Elie's closed-form bulk radial tower Casimirs + L4 T190 tension all integrated.
- Three explicit resolutions (A: reorder Grace mapping; B: T190 exponent = gauge mediator; C: exponent elsewhere) with explicit preference for B + honest scope on disambiguation requiring experimental input (branching ratios, CP phases).
- Resolution B framing is structurally clean: $\text{spinor}^3 \{0,4,6\} = \text{STRUCTURAL channel labels}$; T190 exponent = adjoint Casimir of gauge mediator (same integer 6, different role).
- Concrete L4 v0.2 derivation target explicit (Section 3): derive $(24/\pi^2)^{\wedge\{C_2\}}$ from Bergman kernel integral on adjoint channel of spinor^3 decomposition.
- Refined particle classification table absorbs Grace channel structure into the lepton sector.
- Cal #27 self-applied: “v0.2 is SYNTHESIS, not new derivation.”

- Honest about Resolution A vs B disambiguation requiring experimental input.

Concern requiring explicit handling — two-structures burden flag:

- Resolution B invokes the integer 6 playing two distinct structural roles: (i) intermediate-Casimir label in Grace’s spinor³ channel decomposition AND (ii) gauge-mediator adjoint Casimir in T190 exponent. This parallels the same “one $h^\vee$ doing double duty” pattern Keeper previously calibrated on the generation-mechanism gate (#414 v0.2 reframe — generations and colors both = 3 = $h^\vee$ but require TWO STRUCTURALLY-INDEPENDENT 3-fold structures, not one primary playing two roles).
- Two readings of Resolution B:
 - Reading 1 (weaker): the integer 6 = C_2 appears in both contexts because both contexts naturally involve the B_2 adjoint Casimir — this is structural over-determination, not double-duty. PASS.
 - Reading 2 (stronger structural burden): Resolution B *requires* the two $C_2=6$ instances to be STRUCTURALLY INDEPENDENT (one as spinor³ intermediate label, one as gauge-mediator adjoint Casimir) for the framework to genuinely resolve the tension. Without independence, Resolution B is “one 6 doing double duty” which Keeper has previously rejected as inadequate at the generation gate.
- Recommendation: v0.3 should explicitly state which reading (1 or 2) applies. If Reading 2, the burden parallels the bulk-color burden — need to demonstrate two structurally-independent instances of $C_2=6$ (one geometric/representation-theoretic in the spinor³ channel structure, one dynamic/gauge-mediator in T190). The work is structurally analogous to bulk-color’s color-geometric vs generation-algebraic split (#418).

Additional notes: - N1 (carried from v0.1): V-compound coproduct burden marker — pending engine v0.3 V-compound layer. - N2 (carried from v0.1): ν -eigenstate granularity — addressed via Resolution B framing. - Pair α mapping cross-CI: Resolution B asks Grace to reconsider her original Gen-2 = vector-channel mapping. Grace should explicitly respond (concur with Resolution B, push back, or propose Resolution D).

Tier disposition: - Three-piece synthesis (Section 0): SYNTHESIS (rigorous absorption of Grace + Elie + L4 work) - Resolution B (Section 1): CANDIDATE FRAMEWORK with two-structures-burden flag - Refined decay mechanism (Section 2): FRAMEWORK (consistent with v0.1) - L4 derivation target (Section 3): CONCRETE OPEN — multi-week kernel-integral work - Particle classification table (Section 4): FRAMEWORK with Resolution-B-preferred mapping

Status: PASS at SYNTHESIS; Resolution B preferred but carries two-structures burden requiring v0.3 disambiguation; Pair α cross-CI alignment with Grace pending.

3. Routed handback

- Lyra:
 - Bulk-color v0.6 absorbs C1 (Toeplitz-source citation) + C2 (AdS/CFT demotion) + C3 (SO(3)-decomposition mismatch handling) + C4 ($[T_a, T_b]$ independence verification routed to Elie).
 - Quasi-Eigentone v0.3 disambiguates Resolution B reading (1 or 2); if Reading 2, explicit two-structures argument needed.
- Elie:
 - $[T_a, T_b]$ independence verification on $H^2(S)$ (numerical, Toeplitz operator algebra commutator structure).
 - Kernel-integral computation deriving T190 closed form $(24/\pi^2)^{\{C_2\}}$ from adjoint channel (multi-week, post-affine-pin).
- Grace:
 - Reconsider Pair α mapping with T190 exponent constraint per Lyra Resolution B; concur, push back, or propose Resolution D.
- Cal:
 - Cold-read v0.5 count-match with focus on the SO(3)-decomposition mismatch (3+3+2 vs 3+5 canonical) — is this a substantive structural problem or resolvable via non-canonical embedding?
 - Cold-read v0.2 Resolution B with the two-structures-burden framing applied.

4. Queue status

Saturday batch tier-gates total: 16 items dispositioned across 7 consolidated docs.

Pulling next: - Two-Route Scan correction note absorbing Cal's relation-reduction brake (Cal: 4-route $\rightarrow$ $\sim$ 2 structurally-independent under $C_2 = N_c \cdot \text{rank}$ and $n_C = \text{rank} + N_c$) - Honest-State Ledger v0.4 with Saturday late-morning batch absorption

— Keeper. Two more tier-gates filed; bulk-color v0.5 count-match honest at IDENTIFIED with explicit C3 $SO(3)$ -decomposition concern (Lyra self-flagged); Quasi-Eigentone v0.2 Resolution B PASS at SYNTHESIS with two-structures-burden flag carrying forward. Both items remain ACTIVE multi-week with explicit next-step conditions.